MIL-E-15809C(NAVY)
6 July 1987
SUPERSEDING
MIL-E-15809B(NAVY)
26 October 1972
(See 6.4)

MILITARY SPECIFICATION

EXPANDER, TUBE, CONDENSER AND HEAT EXCHANGERS

This specification is approved for use within the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tube expanders of the self-feeding type for use in expanding admiralty, copper-nickel alloy or other nonferrous alloy tubes in all types of tubular heat exchangers.

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3441
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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Bodies and collars. Bodies, collars and collar sleeves shall be made of heat-treated alloy steel.

3.1.2 Mandrels and rolls. Mandrels and rolls shall be made of tool steel, tempered and hardened to withstand the shocks and wear of regular usage.

3.1.3 Recovered materials. Unless otherwise specified herein, all material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Construction. Expanders shall consist of a mandrel, body, collar, collar sleeve, and three rolls. Unless otherwise specified (see 6.2), expanders shall be righthand feed.

3.2.1 Mandrel. The portion of the mandrel in contact with the rolls shall have a taper of 0.020 to 0.032 inch per inch of mandrel length. To prevent the expander body from remaining in the tube when the mandrel is withdrawn, the mandrel shall have a threaded end with nut, except that mandrels for 1/4-inch and 3/8-inch size expanders may use other means for mandrel retention. The mandrels for 5/8-inch and 3/4-inch size tube expanders shall be provided with either a 3/8-inch square shank or a No. 1 Morse taper shank as specified (see 6.2). The mandrels for 1/4-inch and 3/8-inch size expanders shall be provided with either a 1/4-inch square shank or a 5/16-inch round shank for a Jacobs chuck as specified see (6.2).

3.2.2 Body. The body shall be slotted to contain three taper rolls, set 120 degrees apart, at a lead angle of 1/4 to 2 degrees, and shall have a threaded section on which shall be mounted a collar and sleeve, adjustable in position on the body so as to regulate the depth of penetration of the rolls into the tube, and secured by a lock nut or other positive means.

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3.2.3 Collar. The collar shall be threaded internally to suit the threads on the body. An antifriction type thrust bearing and a sleeve shall bear against the lower end of the body.

3.2.4 Collar sleeve. The collar sleeve shall surround the upper ends of the taper rolls. The collar sleeve of all expanders except the 1/4-inch size shall be provided with a recess 1/16-inch deep (tolerence plus 1/64 inch minus 0) at the lower end. The diameter of the collar sleeve recess shall be not less than 1/32 inch larger than the outside diameter of the tubes for which the expander is intended. The sleeve shall be held in the collar but shall be free to rotate in reference to the collar.

3.2.5 Rolls. The rolls shall have a taper which shall measure 0.010 to 0.016 inch per inch of roll length so as to produce parallel expansion of the tube. The effective length of the rolls shall be such that, together with position adjustment of the collar, the expander will be suitable for rolling the tubes to within 1/8 inch of the inner face of the tube sheet, the thickness of which is specified or to the maximum depth specified (see 6.2). The rolls shall be ground or polished, and shall be rounded at the entrance end to prevent scoring of the tube. The rolls shall be arranged so that they will not fall out of the body unless the expander is disassembled. When the mandrel is inserted to within 1/4 inch of the shoulder, the rolls shall not be body-bound.

3.2.6 Sizes. Expanders shall be furnished in the sizes shown in table I (see 6.2).

TABLE I. Sizes.

Outside diameter of tube	Wall thickness of tube
Inch	Inch
1/4	0.028
1/4	.035
3/8	.049
3/8	.035
3/8	.028
5/8	.065
5/8	.049
5/8	.035
3/4	.065
3/4	.049
3/4	.035

3.3 Spares. A complete set of rolls shall be furnished with each expander.

3.4 Marking. The size and the manufacturer's name shall be indelibly stamped on each expander.

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3.5 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection.

4.2.1 Inspection lot. All assembled expanders or all spare rolls of the same size submitted for inspection at one time shall be considered an inspection lot. Inspection shall include the examination of 4.3 and the inspection of packaging of 4.4.

4.2.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be 1.5 defects per hundred units.

4.3 Examination.

4.3.1 In process examination. Intermediate assemblies which cannot be checked or measured after final assembly shall be examined at various stages of manufacture for conformance to this specification. Whenever a deviation is noted, correction shall be made. Failure to make immediate correction may cause rejection of the affected end items.

4.3.2 Visual and dimensional examination. Each of the assembled expanders shall be examined to determine conformance to this specification regarding dimensions, fits, finishes, security of assembly, and freedom of movement of the parts. Each of the spare rolls shall be examined to determine conformance to this specification regarding dimensions and finishes. Any expander or spare roll which does not conform to this specification shall not be offered for delivery.

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4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservative-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation-packaging, packing, and marking. Tube expanders shall be preserved and packaged level A or C, packed level A, B, or C as specified (see 6.2), and marked in accordance with PPP-P-40.

6. NOTES

6.1 Intended use. The tube expanders covered by this specification are intended for use in all types of tubular heat exchangers.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Expanders other than righthand feed (see 3.2).
(c) For 5/8-inch and 3/4-inch size tube expanders, whether 3/8-inch square shank or No. 1 Morse taper shank is required (see 3.2.1).
(d) For 1/4-inch and 3/8-inch size tube expanders, whether 1/4-inch square shank or 5/16 inch round shank is required (see 3.2.1).
(e) Thickness of tube sheet in which tube is to be expanded or maximum depth of expansion (see 3.2.5).
(f) Size of tube expanders required and wall thickness of tube (see 3.2.6).
(g) Levels of preservation-packaging and packing (see 5.1).

6.3 Subject term (key word) listing.

Condenser
Copper-nickel
Expander, tube
Heat exchanger

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Review activity:
Navy - SH

User activity:
Navy - YD

Preparing activity:
Navy - SH
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-15809C	Expander, Tube, Condenser And Heat Exchangers

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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